Android: Basics on development

Desenvolvimento de Software e Sistemas Móveis (DSSMV)
Licenciatura em Engenharia de Telecomunicações e Informática
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Disclaimer

Material and Slides

Some of the material/slides are adapted from various:

- Presentations found on the internet;
- Books;
- Web sites;
- ...

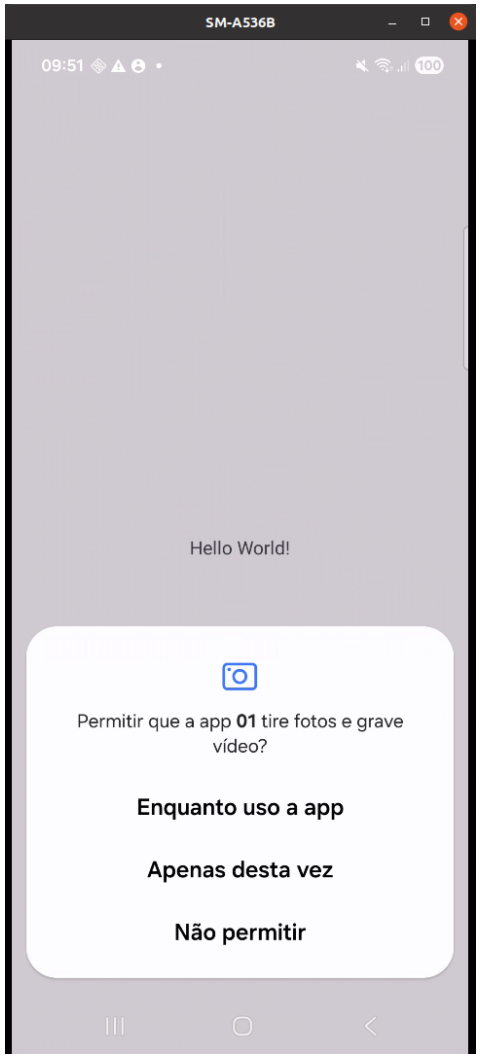
Outline

1 Permissions

2 Networking

Permissions

Multiple Permissions

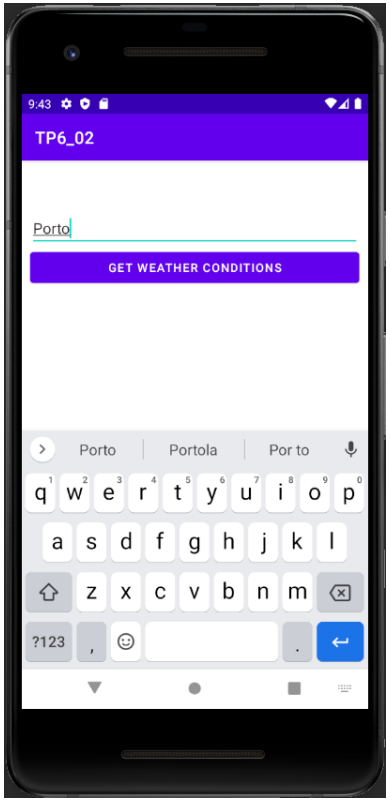


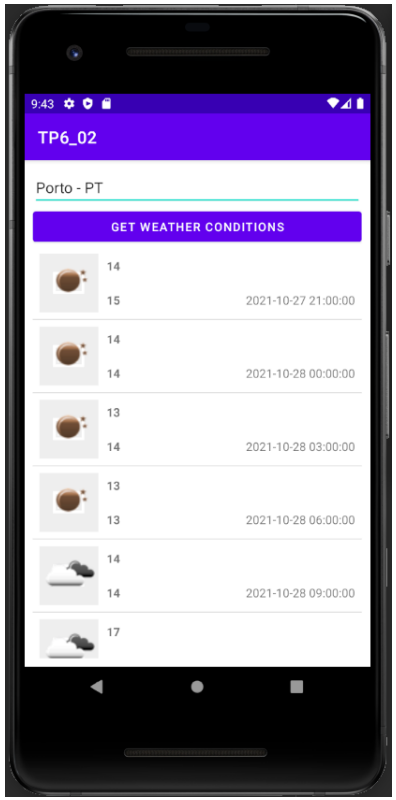
Check: `TP05_01`

Networking

Networking (I)

- Getting data from `http://api.openweathermap.org/`
- Data in JSON format

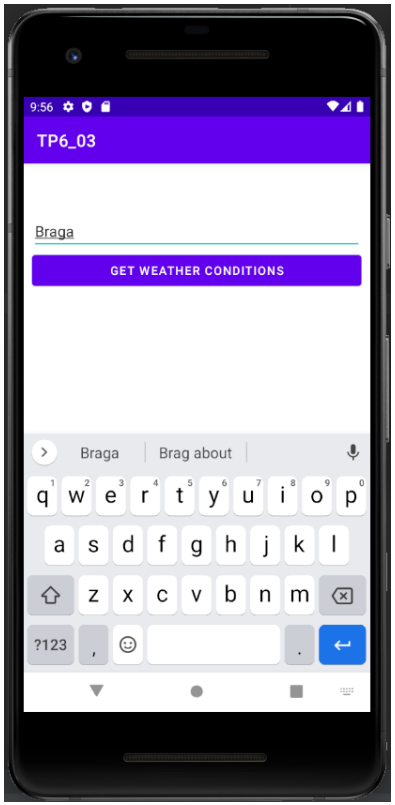


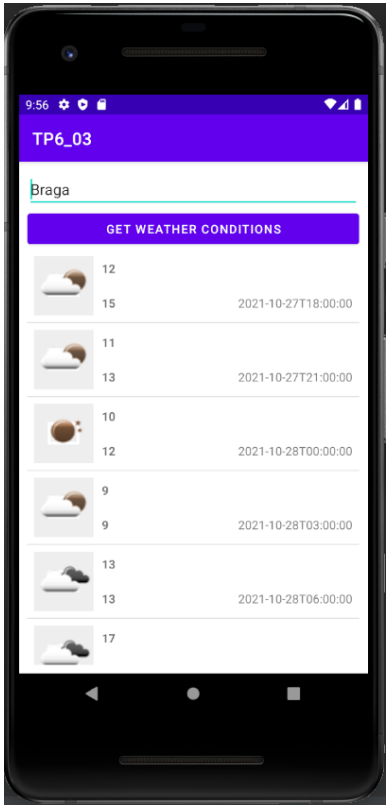


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Networking (II)

- Getting data from `http://api.openweathermap.org/`
- Data in XML format

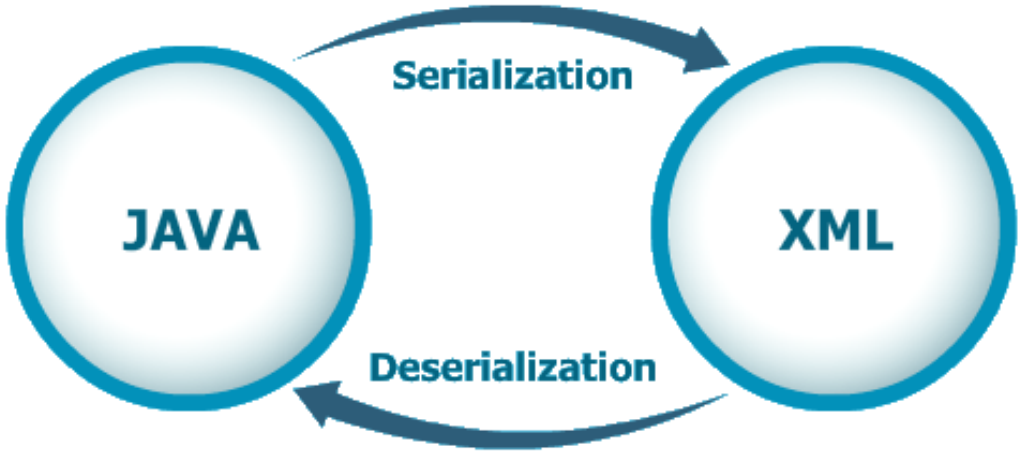




Check: `TP05_03`

Object to/from XML

- **Serialization** is the process of translating Java objects into XML documents
- **Deserialization** is the process of translating XML documents into Java objects .



XML: Example

```
public class User {
  private int id;
  private String name;
  private String profession;
  public User(){}
  public int getId() {
    return id;
  }
  public void setId(int id) {
    this.id = id;
  }
  public String getName() {
    return name;
  }
  public void setName(String name) {
    this.name = name;
  }
  public String getProfession() {
    return profession;
  }
  public void setProfession(String
        profession) {
    this.profession = profession;
  }
}
```

```
<?xml version="1.0" encoding="UTF-8"
      standalone="yes"?>
<users>
  <user>
    <id>1</id>
    <name>Maria</name>
    <profession>Teacher</profession>
  </user>
  <user>
    <id>2</id>
    <name>Pedro</name>
    <profession>Teacher</profession>
  </user>
  <user>
    <id>3</id>
    <name>Joana</name>
    <profession>Teacher</profession>
  </user>
</users>
```

XML: Create a Serializer

```
public String serializer(ArrayList<User> users) {
  StringWriter writer = new StringWriter();
  XmlPullParserFactory factory = XmlPullParserFactory.newInstance();
  XmlSerializer serializer = factory.newSerializer();
  serializer.setOutput(writer);
  serializer.startDocument("UTF-8", true);
  serializer.startTag("", "users");
  for (User user: users){
    serializer.startTag("", "user");
    serializer.startTag("", "id");
    serializer.text(Integer.toString(user.getId()));
    serializer.endTag("", "id");
    serializer.startTag("", "name");
    serializer.text(user.getName());
    serializer.endTag("", "name");
    serializer.startTag("", "profession");
    serializer.text(user.getProfession());
    serializer.endTag("", "profession");
    serializer.endTag("", "user");
  }
  serializer.endTag("", "users");
  serializer.endDocument();

  return writer.toString();
}
```

XML: Create a Parser (Deserializer)

```
public ArrayList<User> parse(InputStream is) {
  ArrayList<User> users= new ArrayList<User>();
  User user = null; String text="";
  XmlPullParserFactory factory = XmlPullParserFactory.newInstance();
  XmlPullParser parser = factory.newPullParser();
  parser.setInput(is, null);
  int eventType = parser.getEventType();
  while (eventType != XmlPullParser.END_DOCUMENT) {
    String tagname = parser.getName();
    switch (eventType) {
    case XmlPullParser.START_TAG:
      if (tagname.equalsIgnoreCase("user")) { user = new User();}break;
    case XmlPullParser.TEXT: text = parser.getText(); break;
    case XmlPullParser.END_TAG:
      if (tagname.equalsIgnoreCase("user")) {
        users.add(user);
      }else if (tagname.equalsIgnoreCase("id")) {
        user.setId(Integer.parseInt(text));
      } else if (tagname.equalsIgnoreCase("name")) {
        user.setName(text);
      } break;
  }
  eventType = parser.next();
  }
  return users;
}
```

JSON: Example

```
{"users":[{"user":{"id":1,"profession":"Teacher","name":"Maria"}},
{"user":{"id":2,"profession":"Teacher","name":"Pedro"}},
{"user":{"id":3,"profession":"Teacher","name":"Joana"}}]}
```

JSON: Create a Serializer

```
public static String serializer (ArrayList<User> data) {
  try {
    JSONObject jsonObj = new JSONObject();
    JSONArray jsonArr = new JSONArray();
    for (User user : data ) {
      JSONObject obj = new JSONObject();
      obj.put("id", user.getId());
      obj.put("name", user.getName());
      obj.put("profession", user.getProfession());
      JSONObject obj1 = new JSONObject();
      obj1.put("user",obj);
      jsonArr.put(obj1);
    }
    jsonObj.put("users", jsonArr);
    return jsonObj.toString();
  }catch(JSONException ex) {
    ex.printStackTrace();
  }
  return null;
}
```

JSON: Create a Parser (Deserializer)

```
public static ArrayList<User> parser(String data) {
  ArrayList<User> array = new ArrayList<>();
  User user;
  try {
    JSONObject jsonObj = new JSONObject(data);
    JSONArray jsonArr = jsonObj.optJSONArray("users");
    for (int i=0; i < jsonArr.length(); i++) {
      JSONObject obj = jsonArr.getJSONObject(i);
      JSONObject obj1 = obj.getJSONObject("user");
      user = new User();
      user.setId(obj1.getInt("id"));
      user.setName(obj1.getString("name"));
      user.setProfession(obj1.getString("profession"));
      array.add(user);
    }
    return array;
  }catch(JSONException ex) {
    ex.printStackTrace();
  }
  return null;
}
```